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Foldable scooter

Abstract

A foldable scooter is provided, including: a deck assembly, including a first deck and a second deck foldably connected and relatively rotatable about a first direction, the first deck including a first lock structure; a steering assembly, connected to the deck assembly; a locking assembly, mounted to the second deck, including a locking member movable between a first position and a second position in a second direction and including a second lock structure extending in the first direction and movable with the locking member; wherein when the locking member is located in the first position, the first and second lock structures are interlocked with each other in a male-female manner so that the deck assembly is non-foldable; when the locking member is located in the second position, the first and second lock structures are unlocked from each other so that the deck assembly is foldable.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to a foldable scooter.

Description of the Prior Art

(2) The scooter is another product of skateboarding besides the skateboard. For easy storage, some conventional scooters are foldable. The conventional foldable scooter generally includes a deck assembly, a first steering assembly and a second steering assembly, one end of the deck assembly is provided with a front wheel and connected to the first steering assembly, and another end of the deck assembly is provided with a rear wheel. In some of the conventional scooters, the first steering assembly and the second steering assembly may be foldably connected to each other.

(3) However, the conventional foldable scooter can just be folded at a position near the deck assembly which is non-foldable, which makes the folded size of the scooter large and requires large space for storing. Besides, the conventional foldable scooter is provided with no self-locking function so that the folded scooter can unfold easily under an external impact.

(4) The present invention is, therefore, arisen to obviate or at least mitigate the above-mentioned disadvantages.

SUMMARY OF THE INVENTION

(5) The main object of the present invention is to provide a foldable scooter which is foldable to reduce the size thereof and easy to carry and store.

(6) To achieve the above and other objects, a foldable scooter is provided, including: a deck assembly, including a first deck and a second deck which are foldably connected to each other and relatively rotatable about a first direction, the first deck including a first lock structure arranged in the first direction; a steering assembly, connected to the deck assembly; a locking assembly, mounted to the second deck, including a locking member movable between a first position and a second position in a second direction transverse to the first direction and including a second lock structure extending in the first direction and movable with the locking member in the second direction; wherein when the locking member is located in the first position, the first lock structure and the second lock structure are interlocked with each other in a male-female manner along the first direction so that the deck assembly is non-foldable; when the locking member is located in the second position, the first lock structure and the second lock structure are unlocked from each other so that the deck assembly is foldable.

(7) The present invention will become more obvious from the following description when taken in connection with the accompanying drawings, which show, for purpose of illustrations only, the preferred embodiment(s) in accordance with the present invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a stereogram of a preferable embodiment of the present invention;

(2) FIG. 2 is a breakdown drawing of a preferable embodiment of the present invention;

(3) FIG. 3 is another breakdown drawing of a preferable embodiment of the present invention;

(4) FIG. 4 is a drawing showing a deck assembly in a locked state according to a preferable embodiment of the present invention;

(5) FIG. 5 is a drawing showing the deck assembly in an unlocked state according to a preferable embodiment of the present invention;

(6) FIG. 6 is a drawing showing a foldable scooter in a folded state according to a preferable embodiment of the present invention;

(7) FIG. 7 is a partial enlargement of FIG. 6; and

(8) FIG. 8 is a stereogram of a locking member according to a preferable embodiment of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(9) Please refer to FIGS. 1 to 8 for a preferable embodiment of the present invention. A foldable scooter 1 of the present invention includes a deck assembly 10, a steering assembly 20 and a

locking assembly **30**.

(10) The deck assembly **10** includes a first deck **11** and a second deck **12** which are foldably connected to each other and relatively rotatable about a first direction **D1**. The first deck **11** includes a first lock structure **111** arranged in the first direction **D1**. The steering assembly **20** is connected to the deck assembly **10**. The locking assembly **30** is mounted to the second deck **12**, and the locking assembly **30** includes a locking member **31** movable between a first position (FIG. **4**) and a second position (FIG. **5**) in a second direction **D2** transverse to the first direction **D1**, and the locking assembly **30** includes a second lock structure **311** extending in the first direction **D1** and movable with the locking member **31** in the second direction **D2**. When the locking member **31** is located in the first position, the first lock structure **111** and the second lock structure **311** are interlocked with each other in a male-female manner along the first direction **D1** so that the deck assembly **10** is non-foldable. When the locking member **31** is located in the second position, the first lock structure **111** and the second lock structure **311** are unlocked from each other so that the deck assembly **10** is foldable. Whereby, the foldable scooter can be folded to reduce the size thereof, and it is easy to carry and store.

(11) The first deck **11** further includes an upper slot **112** and a lower slot **113**, the second deck **12** includes an upper plate **121** and a lower plate **122**, and when the locking member **31** is located in the first position, the upper plate **121** and the lower plate **122** are inserted in the upper slot **112** and the lower slot **113**, respectively.

(12) The locking member **31** further includes a through hole **312** disposed therethrough in the first direction **D1**, and a shaft **40** by which the first deck **11** and the second deck **12** are foldably connected is mounted to the first deck **11** and disposed through the through hole **312**. As such, the shaft **40** is relatively movable along the through hole **312** in the second direction **D2** so that the locking member **31** can be moved to disengage the second lock structure **311** from the first lock structure **111**.

(13) Preferably, a latch pin **32** extending in the first direction **D1** is mounted to the locking member **31**, two sidewalls of the first deck **11** arranged oppositely in the first direction **D1** each include a notch **114**, and when the locking member **31** is located in the first position, the latch pin **32** is engaged in the notches **114** of the two sidewalls of the first deck **11**, which enhances the reliability, stability and structural strength of the locked configuration of the first deck **11** and the second deck **12**. In other embodiment, the latch pin **32** and the locking member **31** may be formed of one piece. A fastener **33** is screwed to the locking member **31** and abutted radially against the latch pin **32** to firmly secure the latch pin **32**. Two sidewalls of the second deck **12** arranged oppositely in the first direction **D1** each include a notch **123**, and when the locking member **31** is located in the second position, the latch pin **32** is located within the notches **123** of the two sidewalls of the second deck **12**. The notches **123** can guide the latch pin **32** and control the travel distance of the latch pin **32** for sufficiently disengaging the second lock structure **311** from the first lock structure **111**.

(14) The steering assembly **20** may be connected to either of the two decks. In this embodiment, the steering assembly **20** is connected to the first deck **11**, the second deck **12** is foldable to be directed toward a lower portion of the steering assembly **20** and to be located between an upper portion of the steering assembly **20** and the first deck **11**, and thus the foldable scooter can be very compact.

(15) Preferably, the steering assembly **20** includes a buckle member **21** (such as a hook or the like), the locking member **31** includes a recess **313**, and the steering assembly **20** is foldable so that the buckle member **21** is releasably buckled in the recess **313** and the second deck **12** is restricted between the steering assembly **20** and the first deck **11**. The second deck **12** is provided with a wheel **124**, and the wheel **124** contacts the steering assembly **20** when the buckle member **21** is buckled in the recess **313**, for conducting a stable folded configuration. The second deck **12** encompasses the locking member **31** and includes an opening **125**, the locking member **31** further includes another recess **314** corresponding to the opening **125**, and in the second direction **D2** the

opening **125** has an extent larger than an extent of the recess **314** so that the locking member **31** can be moved by one or more fingers through the opening **125**.

(16) The locking assembly **30** further includes an elastic member **34** disposed between the second deck **12** and the locking member **31**. When the locking member **31** is located in the first position, the elastic member **34** biases the locking member **31** so that the second lock structure **311** is urged and abutted on the first lock structure **111** to ensure the interlocking of the first lock structure **111** and the second lock structure **311**. When the buckle member **21** is inserted in the recess **313**, the elastic member **34** biases the locking member **31** to urge and restrict the buckle member **21**. In this embodiment, the elastic member **34** is a coil spring, the locking member **31** further includes a C-shaped groove **315** extending in the first direction **D1**, and an end of the coil spring is blocked and engaged in the C-shaped groove **315**. As such, the coil spring can be quickly and reliably connected to the locking member **31**.

(17) The locking member **31** is slidably received within the second deck **12**, one of the first lock structure **111** and the second lock structure **311** includes a shaft **115** extending in the first direction **D1**, and the other of the first lock structure **111** and the second lock structure **311** includes a groove **316** extending in the first direction **D1** within which the shaft **115** is releasably engaged in the second direction **D2**. In this embodiment, the first lock structure **111** includes the shaft **115**, and the second lock structure **311** includes the groove **316**.

(18) Preferably, the first deck **11** further includes a chamber **116**. When the locking member **31** is located in the first position, the first lock structure **111** is located in the chamber **116**, the locking member **31** retractably projects out beyond the second deck **12** so that the second lock structure **311** is inserted in the chamber **116**, and an interior wall of the chamber **116** encompasses an end of the locking member **31**, which enhances the reliability, stability and structural strength of the locked configuration of the first deck **11** and the second deck **12**.

(19) Although particular embodiments of the invention have been described in detail for purposes of illustration, various modifications and enhancements may be made without departing from the spirit and scope of the invention. Accordingly, the invention is not to be limited except as by the appended claims.

Claims

1. A foldable scooter, including: a deck assembly, including a first deck and a second deck which are foldably connected to each other and relatively rotatable about a first direction, the first deck including a first lock structure arranged in the first direction; a steering assembly, connected to the deck assembly; and a locking assembly, mounted to the second deck, including a locking member movable between a first position and a second position in a second direction transverse to the first direction and including a second lock structure extending in the first direction and movable with the locking member in the second direction; wherein when the locking member is located in the first position, the first lock structure and the second lock structure are interlocked with each other in a male-female manner along the first direction so that the deck assembly is non-foldable; when the locking member is located in the second position, the first lock structure and the second lock structure are unlocked from each other so that the deck assembly is foldable; wherein the steering assembly is connected to the first deck, and the second deck is foldable to be directed toward the steering assembly and to be located between the steering assembly and the first deck; wherein the steering assembly includes a buckle member, the locking member includes a recess, and the steering assembly is foldable so that the buckle member is releasably buckled in the recess and the second deck is restricted between the steering assembly and the first deck.

2. The foldable scooter of claim 1, wherein the second deck is provided with a wheel, and the wheel contacts the steering assembly when the buckle member is buckled in the recess.

3. The foldable scooter of claim 1, wherein the locking assembly further includes an elastic

member disposed between the second deck and the locking member; when the buckle member is inserted in the recess, the elastic member biases the locking member to urge and restrict the buckle member.

4. The foldable scooter of claim 1, wherein the locking assembly further includes an elastic member disposed between the second deck and the locking member; when the locking member is located in the first position, the elastic member biases the locking member so that the second lock structure is urged and abutted on the first lock structure.

5. The foldable scooter of claim 1, wherein the first deck further includes an upper slot and a lower slot, the second deck includes an upper plate and a lower plate, and when the locking member is located in the first position, the upper plate and the lower plate are inserted in the upper slot and the lower slot, respectively.

6. The foldable scooter of claim 1, wherein the locking member further includes a through hole disposed therethrough in the first direction, a shaft by which the first deck and the second deck are foldably connected is mounted to the first deck and disposed through the through hole, and the shaft is relatively movable along the through hole in the second direction.

7. The foldable scooter of claim 1, wherein the locking assembly further includes an elastic member which is a coil spring and disposed between the second deck and the locking member, the locking member further includes a C-shaped groove extending in the first direction, an end of the coil spring is blocked and engaged in the C-shaped groove.

8. The foldable scooter of claim 1, wherein the second deck encompasses the locking member and includes an opening, the locking member further includes another recess corresponding to the opening, and in the second direction the opening has an extent larger than an extent of the another recess.

9. A foldable scooter, including: a deck assembly, including a first deck and a second deck which are foldably connected to each other and relatively rotatable about a first direction, the first deck including a first lock structure arranged in the first direction; a steering assembly, connected to the deck assembly; and a locking assembly, mounted to the second deck, including a locking member movable between a first position and a second position in a second direction transverse to the first direction and including a second lock structure extending in the first direction and movable with the locking member in the second direction; wherein when the locking member is located in the first position, the first lock structure and the second lock structure are interlocked with each other in a male-female manner along the first direction so that the deck assembly is non-foldable; when the locking member is located in the second position, the first lock structure and the second lock structure are unlocked from each other so that the deck assembly is foldable; wherein the locking member is slidably received within the second deck, one of the first lock structure and the second lock structure includes a shaft extending in the first direction, and the other of the first lock structure and the second lock structure includes a groove extending in the first direction within which the shaft is releasably engaged in the second direction.

10. The foldable scooter of claim 9, wherein the first deck further includes a chamber, the first lock structure is located in the chamber, the locking member retractably projects out beyond the second deck so that the second lock structure is inserted in the chamber, and an interior wall of the chamber encompasses an end of the locking member.

11. A foldable scooter, including: a deck assembly, including a first deck and a second deck which are foldably connected to each other and relatively rotatable about a first direction, the first deck including a first lock structure arranged in the first direction; a steering assembly, connected to the deck assembly; and a locking assembly, mounted to the second deck, including a locking member movable between a first position and a second position in a second direction transverse to the first direction and including a second lock structure extending in the first direction and movable with the locking member in the second direction; wherein when the locking member is located in the first position, the first lock structure and the second lock structure are interlocked with each other

in a male-female manner along the first direction so that the deck assembly is non-foldable; when the locking member is located in the second position, the first lock structure and the second lock structure are unlocked from each other so that the deck assembly is foldable; wherein a latch pin extending in the first direction is mounted to the locking member, two sidewalls of the first deck arranged oppositely in the first direction each include a notch, and when the locking member is located in the first position, the latch pin is engaged in the notches of the two sidewalls of the first deck.

12. The foldable scooter of claim 11, wherein a fastener is screwed to the locking member and abutted radially against the latch pin.

13. The foldable scooter of claim 11, wherein two sidewalls of the second deck arranged oppositely in the first direction each include a notch, and when the locking member is located in the second position, the latch pin is located within the notches of the two sidewalls of the second deck.
